

# Complete Mission Flight Plan

April 22, 2026

*"Eight human beings. One trajectory. No abort. No rescue. No second chances."*

|            |                                                              |
|------------|--------------------------------------------------------------|
| Mission    | Oblivion I — Crewed Mars Orbital Insertion                   |
| Vehicle    | ARES-1 — 150 m x 15 m — Vertical Integrated                  |
| Crew       | 8 Astronauts — Two teams of four on alternating watch cycles |
| Duration   | ~500 to 580 days total mission                               |
| Phases     | 10 mission phases — Earth to Mars and back                   |
| Propulsion | NTP engine block — ignites from orbit only — Isp ~900 s      |
| Refueling  | Phase 2 (Earth orbit) + Phase 5 (Low Mars Orbit depot)       |
| Delta-V    | ~5,900 to 6,200 m/s total                                    |
| Return     | Capsule separation + skip reentry at 12-14 km/s + splashdown |

## Mission Overview

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Oblivion I is the first mission of the Horizon Program and the most ambitious human spaceflight mission ever attempted. Eight crew members will travel from Earth to Mars, enter orbit around another planet for the first time in human history, conduct intensive scientific operations, and return safely to Earth across approximately 500 to 580 days of deep interplanetary space travel. The mission is divided into ten phases, each with distinct operational characteristics and risk profiles.

The mission is made possible by three foundational architectural decisions. First, ARES-1's NTP engine ignites exclusively from orbit — meaning the vehicle's entire delta-V budget of approximately 6,000 m/s is directed at the mission itself with nothing wasted fighting Earth's atmosphere. Second, a two-point refueling architecture eliminates the need to carry Earth-return propellant from Earth — it is pre-positioned at Mars. Third, the orbital assembly architecture allows a vehicle of unprecedented size to be constructed in space from manageable launch segments.

### Phase 1 — Orbital Assembly at Earth Staging Orbit

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ARES-1 is built module by module in Earth staging orbit at approximately 400 to 600 km altitude. Oblivion XXX flies a minimum of seven primary missions — one per major module — plus dedicated tanker missions and the crew launch. Canada's robotic assembly systems connect each module as it arrives. Ground teams verify every connection, seal, and electrical interface before the next module launches. The assembled vehicle is structurally complete but carries no propellant at this stage. Assembly takes weeks to months depending on launch cadence.

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|-----------------------------|----------------------------------------------------|
| <b>Duration</b>             | Weeks to months                                    |
| <b>Key Event</b>            | ARES-1 built in orbit from 7 modules               |
| <b>Vehicle State</b>        | Dry — no propellant                                |
| <b>Oblivion XXX Flights</b> | 7+ module launches + tanker missions + crew launch |

### Phase 2 — Earth Orbital Refueling

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With ARES-1 fully assembled and checked out, Oblivion XXX flies dedicated tanker missions delivering liquid hydrogen to the Earth orbital depot alongside ARES-1. The depot maintains propellant at minus 253 degrees Celsius using active cryocoolers powered by its own solar arrays. Once the full outbound propellant load is confirmed across all tanker deliveries, the depot transfers propellant to ARES-1's NTP tanks. The crew launches separately on a dedicated human-rated Oblivion XXX flight and docks with the assembled vehicle. Final systems checks run across all five core systems. The Flight Surgeon clears all eight crew members medically. Mission control gives the final go. The program has committed.

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|----------------------------|------------------------------------------------------|
| <b>Duration</b>            | Days                                                 |
| <b>Key Event</b>           | Full outbound propellant transferred and crew boards |
| <b>Critical Check</b>      | All 5 systems nominal — crew medically cleared       |
| <b>Point of Commitment</b> | Mission control final go/no-go                       |

### Phase 3 — Trans-Mars Injection Burn

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ARES-1's integrated NTP engine block ignites for the trans-Mars injection burn — the point of no return. The reactor reaches operating temperature, liquid hydrogen flows through the core, and the engine produces sustained thrust at approximately 900 seconds of specific impulse. The burn lasts 20 to 40 minutes. When the engine cuts off, ARES-1 is on a heliocentric transfer trajectory. Earth is behind them. There is no abort back to Earth orbit. The crew is committed to the trajectory for the duration of the mission.

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|---------------------|----------------------------------------------|
| <b>Duration</b>     | 20 to 40 minutes                             |
| <b>Delta-V</b>      | ~3,600 m/s                                   |
| <b>Key Event</b>    | Point of no return — no Earth abort possible |
| <b>Engine State</b> | Full thrust — Isp ~900 s                     |

### Phase 4 — Outbound Transit

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The outbound journey covers approximately 200 to 260 days. Two teams of four operate on alternating eight-hour watch cycles ensuring ARES-1 is never unmonitored. The Flight Engineer monitors the NTP system and power distribution continuously. The Systems Engineer watches avionics and navigation, verifying optical position fixes daily. The Mechanical Engineer conducts regular structural inspections of all accessible inter-module joints and pressure seals. The Life Support Specialist manages the closed-loop ECLSS serving eight crew members. The Flight Surgeon runs daily physical and psychological health assessments. The Mission Specialist conducts deep space science observations. All eight crew members maintain the mandatory two-hour daily exercise regime. Communication delay increases steadily, reaching 20 minutes one-way at maximum Mars distance.

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| <b>Duration</b>            | ~200 to 260 days                                              |
| <b>Key Event</b>           | Deep space transit — continuous watch rotations               |
| <b>Communication Delay</b> | Growing to 20 minutes one-way at maximum distance             |
| <b>Critical Routine</b>    | 2-hour daily exercise regime — mandatory medical prescription |

### Phase 5 — Low Mars Orbit Depot Rendezvous and Refueling

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Approximately 10 to 14 days before Mars orbital insertion, ARES-1 begins the autonomous rendezvous sequence with the pre-positioned LMO depot. The depot was launched months earlier on a dedicated precursor mission and has been waiting in a carefully selected Mars orbit. The rendezvous is fully autonomous — at 20-minute communication delay, ground control cannot participate in real-time. ARES-1's avionics execute the approach using optical navigation, closing distance through a series of small correction burns. Once docked, the Earth return propellant transfers from the depot to ARES-1's NTP tanks over several hours. Two independent depots are pre-positioned in separate Mars orbits — if the primary has malfunctioned, the secondary covers. After refueling and undocking, both depots remain in Mars orbit as permanent infrastructure assets available to all future Horizon Program missions.

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|-------------------|--------------------------------------------------------------------|
| <b>Duration</b>   | ~3 to 5 days                                                       |
| <b>Key Event</b>  | Earth return propellant loaded at Mars depot                       |
| <b>Autonomy</b>   | Fully autonomous rendezvous — no real-time ground support possible |
| <b>Redundancy</b> | Two independent depots — primary and secondary                     |

## Phase 6 — Mars Orbital Insertion Burn

Fully fueled for the complete return journey, ARES-1 fires its NTP engine block for the Mars Orbital Insertion burn, decelerating from the heliocentric transfer trajectory into Low Mars Orbit. This burn requires approximately 900 m/s of delta-V and lasts 20 to 40 minutes. This is the moment that transforms Oblivion I from an interplanetary voyage into the most historic human spaceflight achievement since Apollo — eight human beings entering orbit around another planet for the first time in history. Mars fills the observation windows of the crew capsule and living room. Surface features are visible. The crew is in orbit around Mars.

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|----------------------|------------------------------------------|
| <b>Duration</b>      | 20 to 40 minutes                         |
| <b>Delta-V</b>       | ~900 m/s                                 |
| <b>Key Event</b>     | First humans to orbit another planet     |
| <b>Vehicle State</b> | Fully fueled for complete return journey |

## Phase 7 — Mars Orbital Science Operations

The duration of Mars orbital operations is to be determined during the design phase, pending life support consumables analysis and propellant margin calculations. During this phase the Mission Specialist directs the full science program from Low Mars Orbit. High-resolution multispectral surface imaging produces datasets orders of magnitude richer than any robotic orbiter. Atmospheric probes are deployed into the upper Martian atmosphere with targeting precision impossible from a flyby trajectory. The magnetometer boom characterises the Martian magnetosphere at close range. Fourteen CubeSat orbiters are released in two precisely timed salvos into calculated orbital configurations — these CubeSats remain at Mars indefinitely after the crew departs, extending the mission's scientific legacy for years. The Flight Surgeon actively manages crew psychological health during the emotionally intense Mars operations phase.

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|----------------------------|---------------------------------------------------------------------|
| <b>Duration</b>            | TBD — pending design phase consumables analysis                     |
| <b>Key Event</b>           | Full orbital science program — probe deployments — CubeSat release  |
| <b>Science Instruments</b> | Multispectral imager, atmospheric probes, magnetometer, 14 CubeSats |
| <b>Legacy</b>              | CubeSats remain in Mars orbit indefinitely                          |

## Phase 8 — Trans-Earth Injection Burn

With science operations complete and all crew aboard the vehicle, ARES-1 fires its NTP engine block for the Trans-Earth Injection burn, departing Mars orbit and committing to the return

trajectory. The propellant for this burn was loaded at the LMO depot in Phase 5 — it did not travel from Earth. This is the architectural decision that makes the entire mission feasible within a reasonable total vehicle mass. Engine cutoff. Mars recedes in the observation windows. The crew takes their last look at the planet they orbited. Then they get back to work. The return journey covers approximately 200 to 260 days.

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| <b>Duration</b>               | 20 to 40 minutes                                           |
| <b>Delta-V</b>                | ~1,100 m/s (from depot propellant — not from Earth)        |
| <b>Key Event</b>              | Departure from Mars orbit — homeward committed             |
| <b>Emotional Significance</b> | Last view of Mars — crew adjustment to return phase begins |

## Phase 9 — Return Transit

The return transit carries a psychological character distinct from the outbound leg. After 300-plus days in space and the emotional intensity of Mars orbital operations, the return phase is when psychological issues peak in analog long-duration missions. The Flight Surgeon's role during this phase is arguably the most demanding of the entire mission. Watch schedules, exercise regimes, private quarters in the 30-metre living room section, the high-bandwidth communications link home, and the social structure of the eight-person crew are all active life support systems during the return. The Mechanical Engineer conducts pre-separation inspection and certification of the crew capsule beginning approximately 30 days before Earth arrival. As Earth grows from a point of light to a visible disk, the crew begins the formal capsule preparation timeline.

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|----------------------|-----------------------------------------------------------------------|
| <b>Duration</b>      | ~200 to 260 days                                                      |
| <b>Key Event</b>     | Return journey — capsule separation preparation                       |
| <b>Critical Risk</b> | Psychological stress peaks — Flight Surgeon most demanding phase      |
| <b>Capsule Prep</b>  | Formal pre-separation inspection begins ~30 days before Earth arrival |

## Phase 10 — Capsule Separation, Skip Reentry, and Splashdown

Approximately three days before Earth arrival, ARES-1 is moving at interplanetary return velocity of 12 to 14 kilometres per second. The crew transfers essential items into the crew capsule and the Commander initiates the separation sequence. The biconic lifting-body crew capsule detaches from the main ARES-1 vehicle. The main vehicle — service module, NTP engine block, and all remaining systems — continues on its trajectory toward a high Earth parking orbit for potential future reuse, or a solar disposal trajectory pending vehicle condition assessment after 550 days in deep space. The crew capsule enters the upper atmosphere at a precisely calculated angle. The skip reentry bleeds off interplanetary velocity across two atmospheric passes. The plasma sheath during peak heating blacks out communications for several minutes. At 8,000 metres drogue parachutes deploy. At 3,000 metres three main parachutes deploy. The recovery fleet closes in. Splashdown. The hatch opens. Eight human beings climb out into daylight for the first time in approximately 550 days. They have orbited Mars. They have come home.

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| <b>Duration</b> | ~30 minutes reentry + recovery |
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| <b>Delta-V</b>       | Atmospheric braking — no propulsion                        |
| <b>Reentry Speed</b> | 12 to 14 km/s — skip reentry across two atmospheric passes |
| <b>Key Event</b>     | Mission complete — all eight crew recovered alive          |

## Mission Timeline Summary

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| Phase                       | Duration      | Key Event                        | Delta-V          |
|-----------------------------|---------------|----------------------------------|------------------|
| 1 — Orbital Assembly        | Weeks-months  | ARES-1 built from 7 modules      | —                |
| 2 — Earth Refueling         | Days          | Propellant loaded — crew boards  | —                |
| 3 — Trans-Mars Injection    | 20-40 min     | Point of no return               | ~3,600 m/s       |
| 4 — Outbound Transit        | ~200-260 days | Deep space cruise                | ~200 m/s corr.   |
| 5 — LMO Depot Rendezvous    | ~3-5 days     | Earth return propellant loaded   | ~50 m/s          |
| 6 — Mars Orbital Insertion  | 20-40 min     | First humans to orbit Mars       | ~900 m/s         |
| 7 — Mars Orbital Operations | TBD           | Full science program             | ~200 m/s         |
| 8 — Trans-Earth Injection   | 20-40 min     | Departure from Mars              | ~1,100 m/s       |
| 9 — Return Transit          | ~200-260 days | Return cruise — capsule prep     | ~100 m/s         |
| 10 — Reentry & Splashdown   | ~30 min       | Crew recovery — mission complete | Aerobraking      |
| TOTAL                       | ~500-580 days | All 8 crew home alive            | ~5,900-6,200 m/s |

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**500 to 580 days. 800 million kilometres. One vehicle. Eight people. One chance.**

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*End of Mission Plan — Oblivion I — April 22, 2026 Next revision triggered by: Design Phase Completion*